

ECG and PPG beat-detection pipeline

parallel branches converging on shared window screening and feature extraction

Raw acquisition

AD8232 3-lead ECG + 5 × MAX30102 PPG (finger, wrist, earlobe, temple, shoulder)

ECG

Polarity orientation

flip if lead inverted
(robust P1/P99; ≥ 1.2 asymmetry guard)

Relative prominence threshold

peak stands out by
 $0.35 \times (P99 - P40)$ of oriented signal

Refractory constraint

0.28 s minimum RR
(≈ 214 bpm cap; rejects T-wave)

RR interval series

PPG

Band-pass filter

0.5–8.0 Hz zero-phase Butterworth (2nd order)
(after uniform resample + outlier scrub)

TERMA moving averages

$MA_{peak} = 111$ ms • $MA_{beat} = 667$ ms

Block-of-interest thresholding

keep runs where $MA_{peak} > MA_{beat} + \beta \cdot \text{mean}$;
reject blocks narrower than 111 ms

Doublet filter

drop peaks $< 0.6 \times$ median PPI,
keep the taller of each close pair

PP interval series

Window screening — physiological plausibility (Orphanidou 2015)

non-overlapping 10 s windows, ≥ 4 intervals each; all three must hold:

- ① $40 \leq \text{HR} \leq 180$ bpm ② no inter-beat gap > 3 s ③ max/min interval ratio < 2.2

Feature extraction

NN cleaning (300–2000 ms gate + Karlsson $\pm 20\%$) → RR-PPI nearest-neighbour matching
agreement: CCC, ICC(A,1), Bland-Altman (bias, LoA), RMSE, MAE HRV: SDNN, LF/HF